5 EXAM PRACTICE

1 A saturated hydrocarbon has the molecular formula C_4H_8.

Which is a possible name for this hydrocarbon?

A butane

B butene

C cyclobutane

D methylpropane [1]

(Total for Question 1 = 1 mark)

2 Which statement about the bonding in a propene molecule is correct?

A the C–C bond is a pi bond only

B the C=C bond is a pi bond only

C the C=C bond is a sigma bond only

D the C–H bond is a sigma bond only [1]

(Total for Question 2 = 1 mark)

3 What is the IUPAC name for $(CH_3)_2C{=}CHCH_2CH_3$?

A 2-methyl-3-ethylprop-2-ene

B 2-methylpent-2-ene

C 2,2-dimethylbut-2-ene

D 4-methylpent-3-ene [1]

(Total for Question 3 = 1 mark)

4 How many different compounds can be represented by the formula C_3H_6?

A 1

B 2

C 3

D 4 [1]

(Total for Question 4 = 1 mark)

5 An equation for the reaction of an alkene is

$CH_3CH_2CH{=}CH_2 + H_2O \rightarrow CH_3CH_2CH(OH)CH_3$

Which is a correct name for this reaction?

A addition

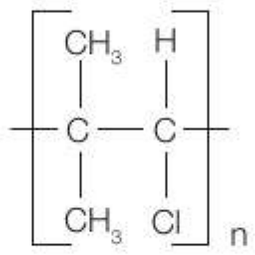
B hydrogenation

C redox

D substitution [1]

(Total for Question 5 = 1 mark)

6 A polymer can be represented by this repeat unit:



What is the name of the monomer that can be used to form this polymer?

A 1-chloro-1,2-dimethylethene

B 1-chloro-2-methylpropene

C 2-chlorobut-2-ene

D 2,2-dimethylchloroethene [1]

(Total for Question 6 = 1 mark)

7 But-1-ene can be produced by the cracking of alkanes.

(a) Draw a dot and cross diagram to show the bonds in a molecule of but-1-ene. [2]

(b) Write an equation for the reaction in which a molecule of dodecane ($C_{12}H_{26}$) is cracked to form two molecules of but-1-ene and one molecule of a saturated hydrocarbon. [1]

(c) A polymer can be made from the monomers ethene and but-1-ene.

Draw the two possible repeat units of the polymer formed from one molecule of each monomer. [2]

(d) But-1-ene can be converted into butane-1,2-diol, $CH_2(OH)CH(OH)CH_2CH_3$

State the reagents needed for this conversion, and the colour change that occurs. [3]

(e) But-1-ene can be distinguished from butane using bromine.

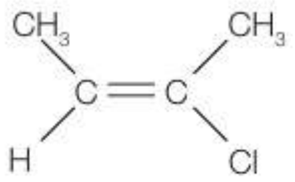
Draw the displayed formula of the organic product formed in the reaction between but-1-ene and bromine.

State the name of this product. [2]

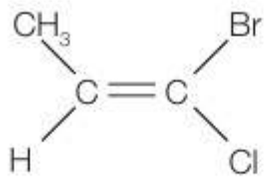
(f) Write the mechanism for the formation of the major product of the reaction between but-1-ene and hydrogen bromide. Include relevant dipoles. [4]

(Total for Question 7 = 14 marks)

8 The structural formulae of two unsaturated compounds, A and B, are shown.



Compound A



Compound B

(a) Draw the formula of a structural isomer of compound A and deduce its name. [2]

(b) Draw the formula of a geometric isomer of compound A and state its name using the *cis-trans* notation. [2]

(c) Deduce the name of compound B, using the *E-Z* notation. [1]

(d) Compound B reacts with chlorine in an addition reaction.
Draw the displayed formula of the product and state its name. [2]

(Total for Question 8 = 7 marks)

9 Ethane-1,2-diol is a product of these reactions.

Reaction 1 $C_2H_4 + [O] + H_2O \rightarrow C_2H_6O_2$

Reaction 2 $C_2H_4Cl_2 + 2NaOH \rightarrow C_2H_6O_2 + 2NaCl$

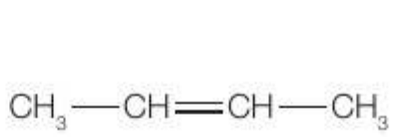
(a) Using Reaction 1, 34.5 kg of ethene were converted into 57.6 kg of ethane-1,2-diol.
Calculate the percentage yield in this conversion. [3]

(b) Calculate the atom economy of Reaction 2. [3]

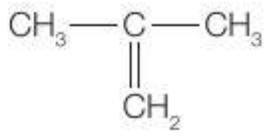
(c) Give two reasons why Reaction 1 is more likely to be used to manufacture ethane-1,2-diol. [2]

(Total for Question 9 = 8 marks)

10 The structures of two alkenes are shown.

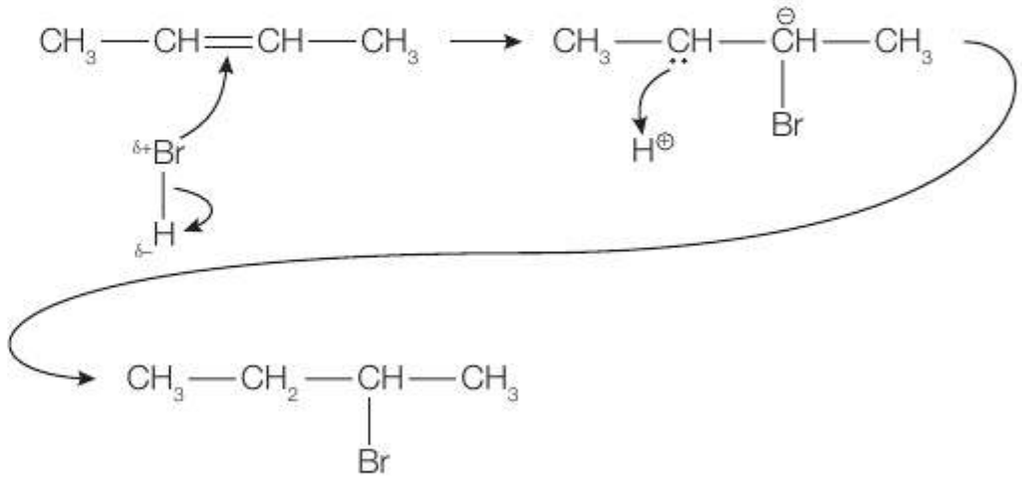


A



B

(a) A student wrote this mechanism for the reaction between alkene **A** and hydrogen bromide.



Identify **six** mistakes in the mechanism. [6]

(b) Draw the two possible structures for the carbocation formed in the reaction between alkene **B** and hydrogen chloride.
Explain which carbocation is the more stable of the two.
Give the name of the major product of the reaction. [5]

(Total for Question 10 = 11 marks)